MCQ MASTERCLASS: SMART MATHS

Quantitative Aptitude & Arithmetic Shortcuts - 2026 Edition

🧠 What this Chapter is REALLY About

Think of this chapter as **“numbers behaving smartly in real life”**:

- **Ratios** → comparing things
- **Percentages** → parts out of 100
- **Interest** → money growing 🌱
- **Profit/Loss** → shopkeeper math
- **Data numbers** → quick calculations

👉 **MCQs are NOT about long solving. They test recognition + shortcut thinking.**

⚡ GOLDEN RULE (Most Important)

✅ **Never solve fully first.**

➡️ **Estimate → Eliminate → Then calculate.**

70% of MCQs can be solved without full calculation.

🚀 SUPER FAST MCQ TRICKS

1️⃣ Percentage → Convert to FRACTION instantly

Percentage	Fraction Trick
50%	1/2
25%	1/4
20%	1/5
12.5%	1/8
33.33%	1/3
66.66%	2/3

Example: 20% of 450 = 1/5 × 450 = **90** (no calculator needed)

EXAM HACK If options are far apart → mental fraction method wins.

2️⃣ Profit & Loss MAGIC FORMULA

Instead of remembering many formulas, remember **ONE**:

```
Profit% = (Profit / CP) × 100
```

Fast Trick: If Profit = 20%, Selling Price = 120% of CP. So, **SP = 1.2 × CP**. No long solving needed.

3️⃣ Successive Percentage Trick 🔥

If increase = a% then decrease = b%, use:

```
Net% = a + b + (ab / 100)
```

⚠️ Add sign carefully (+/−).

Example: +20% then -10% → 20 - 10 - (20×10)/100 = 10 - 2 = **8% increase**. (Done in 5 seconds)

4️⃣ Simple Interest vs Compound Interest Shortcut

Key Observation Trick: If time = 2 years,

```
CI - SI = (P × R²) / 100²
```

No full compound formula needed 😎

5️⃣ Ratio Questions — SMART SCALING

Example: Ratio = 3 : 5, Total = 80.
Add ratio parts: 3 + 5 = 8.
One part = 80 ÷ 8 = 10.
Numbers = 30 and 50 ✅

6️⃣ Option Substitution Trick (Topper Secret ⭐)

If options are numbers, put option into question instead of solving.
Example: “Number increased by 20% becomes 72”
Options: 50, 60, 65, 70
Check quickly: **60 + 20% = 60 + 12 = 72** ✅ (Solved in 3 seconds)

7️⃣ Unit Digit Trick (Huge Time Saver)

If options differ at last digit: **23 × 47** → 3 × 7 = 21 → last digit **1**. Only option ending with 1 is correct.

8️⃣ Approximation Method (Exam King 👑)

Round numbers slightly: **49 × 19 ≈ 50 × 20 = 1000**. Choose closest option.

9️⃣ Interest Growth Shortcut

If rate = 10%, each year multiply by 1.1.
After 2 years: **P × 1.1 × 1.1**. Super fast mental math.

🔟 Elimination Rule (Most Powerful)

❌ Remove impossible options: **negative value? too big? too small?** Often reduces 4 options to 2 instantly. Probability of correct answer doubles!

🧩 5-SECOND MCQ STRATEGY

1. Read question
2. Guess approximate answer
3. Eliminate wrong options

4. Apply shortcut
5. Calculate only if needed

🎯 TOPPER EXAM MANTRA

👉 **MCQs reward speed thinking, not long maths.**

If solution taking > 20 sec → you're doing it wrong.

By Thunderstudy

🧠 MATRICES MCQ — SUPER FAST TRICKS

Class 12 Applied Maths — Smart Solving Guide

Think of a matrix as a number table. MCQs mostly test patterns, not long solving. Goal = avoid calculation whenever possible.

⚡ TRICK 1: Order (Size) Check — Instant Elimination

Matrix order = Rows × Columns

If question asks A + B, addition is possible **ONLY** when orders are the same. If options include a different order → eliminate instantly. 50% of questions are solved this way.

⚡ TRICK 2: Matrix Multiplication Rule (Golden Rule)

Rule: (m × n) × (n × p) = (m × p)

Memory Trick: **Inside numbers must match.**

Example: 2×3 × 3×4 ✅ Possible | 2×3 × 2×4 ❌ Impossible.

⚡ TRICK 3: Identity Matrix Magic (BIGGEST TIME SAVER)

Identity Matrix (I) behaves like the number 1.

```
AI = IA = A
```

If an option shows AI, IA, or A → they are **ALL THE SAME**. No multiplication required 😎

⚡ TRICK 4: Zero Matrix Shortcut

Zero matrix = all elements 0. Multiplication with a zero matrix results in a zero matrix. Immediate answer!

⚡ TRICK 5: Transpose Trick (Swap Swap!)

Transpose = rows ⇔ columns swap. Formula: **$(A^T)^T = A$**

Formula: **$(A+B)^T = A^T + B^T$**

EXAM TRAP: $(AB)^T = B^TA^T$ (Order Reverses!)

Remember: **Transpose = Reverse Order**

⚡ TRICK 6: Symmetric vs Skew-Symmetric (1-Second Check)

Symmetric: $A^T = A$ (Mirror same)

Skew-Symmetric: $A^T = -A$

⭐ **SUPER SHORTCUT: All diagonal elements = 0 in a skew-symmetric matrix. No need for a full transpose check!**

⚡ TRICK 7: Equality of Matrices (FREE MARKS)

Two matrices are equal only if they have the **same order** and **corresponding elements are equal.** Compare just one unequal element from the options to eliminate wrong choices immediately.

⚡ TRICK 8: Scalar Multiplication

If matrix multiplied by number k, every element × k. **Hack:** Check only one element instead of the full matrix to save time.

⚡ TRICK 9: Determinant-Type Pattern Guess

If rows or columns are proportional (e.g., Row 2 = 2 × Row 1), the **determinant = 0**. No calculation needed.

⚡ TRICK 10: Option Substitution (TOPPER SECRET)

Instead of solving: Put the option value into the equation. Check which satisfies the condition. MCQs become 5-second questions.

🚀 5-SECOND EXAM STRATEGY

1. Check order
2. Look for identity/zero matrix
3. Check transpose rules
4. Eliminate impossible options
5. Substitute options if needed

Solve mentally before writing anything!

🧩 MEMORY CODE (Revise Before Exam)

O I Z T S E S D O

Order | Identity | Zero | Transpose | Symmetric | Equality | Scalar | Determinant Pattern | Option Check

---

By Thunderstudy

⚡ DETERMINANTS MCQ — SUPER FAST TRICKS

Class 12 Applied Maths — By Thunderstudy

Think of a determinant like a machine that gives one number from rows & columns. Your goal in MCQs = avoid full solving 😄

🧠 TRICK 1: Two Rows/Columns Same ⇒ Answer = 0

👉 If any two rows OR columns are identical, determinant is **ZERO**.

```
| 2 3 |
| 2 3 | = 0
```

⭐ **MCQ shortcut:** Scan rows first before solving.

🧠 TRICK 2: One Row/Column All Zeros ⇒ Answer = 0

If any row or column has only zeros, Determinant = 0 instantly.

```
| 1 2 3 |
| 0 0 0 |
| 4 5 6 | = 0
```

🧠 TRICK 3: Proportional Rows ⇒ Determinant = 0

If one row is a multiple of another (e.g., $Row_2 = 2 \times Row_1$), Det = 0.

💡 Look for ratios quickly: **2→4, 3→6, 5→10**. Solved in 3 seconds!

🧠 TRICK 4: Swap Rows ⇒ Sign Changes Only

If rows are interchanged, determinant becomes negative: $|A| \rightarrow -|A|$

⭐ Useful when options differ only by sign.

🧠 TRICK 5: Triangle Shape = Multiply Diagonal

If matrix is Upper triangular OR Lower triangular, Determinant = diagonal multiplication.

```
| 2 1 4 |
| 0 5 3 |
| 0 0 7 | = 2 × 5 × 7 = 70
```

🚀 **NO expansion needed.**

🧠 TRICK 6: Factor Common Number OUT

If a row has a common factor, take it out. Takes numbers smaller → faster solving.

🧠 TRICK 7: 2×2 Determinant = Cross Multiply Rule

MOST IMPORTANT 🔥

Formula: `ad - bc` (Main diagonal − Other diagonal)

🧠 TRICK 8: Add/Subtract Rows (Magic Trick ⭐)

Operations like $\mathbf{R_2 \rightarrow R_2 - R_1}$ or $\mathbf{C_2 \rightarrow C_2 + C_1}$ DO NOT change the value. Purpose: Create zeros for easier solving.

🧠 TRICK 9: Identity Matrix Shortcut

If matrix is Identity (diagonal 1s, others 0), Determinant = **1**.

🧠 TRICK 10: Parameter MCQs (x, k etc.)

If determinant = 0, look for proportional rows after substitution. Most parameter questions secretly test this.

🚀 5-SECOND EXAM STRATEGY

1. Step 1 — Check identical rows
2. Step 2 — Check zeros row
3. Step 3 — Check proportional rows
4. Step 4 — Check triangular form
5. Step 5 — Use ad−bc or diagonal multiply

⚡ 70% MCQs finish without full solving.

🎯 GOLDEN MEMORY LINE (Revise Before Exam)

"Same / Zero / Multiple → Determinant Zero
Triangle → Multiply diagonal | 2×2 → ad − bc"

By Thunderstudy

🧠 SIMULTANEOUS LINEAR EQUATIONS

Class 12 Applied Maths — Fast MCQ Tricks — By Thunderstudy

Think of equations like two roads crossing. The solution = where both roads meet. Most MCQs don't need full solving — just smart checking!

✅ 1️⃣ The CROSS-MULTIPLICATION MAGIC

Given: $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$

⚡ Direct Formula (No Long Steps):

```
x = (b₁c₂ - b₂c₁) / (a₁b₂ - a₂b₁)
y = (c₁a₂ - c₂a₁) / (a₁b₂ - a₂b₁)
```

Memory Trick: Write in circle order x → b c, y → c a, den. → a b. Rotate letters clockwise 🔄.

✅ 2️⃣ DETERMINANT TEST (MCQ TRICK)

Calculate: $\mathbf{D = a_1b_2 - a_2b_1}$

Condition	Meaning
$D \neq 0$	Unique solution ✅
$D = 0$ (Ratios equal)	Infinite solutions ♾
$D = 0$ (Ratios unequal)	No solution ❌

🧠 Ratio Trick:

Check a_1/a_2, b_1/b_2, c_1/c_2. All equal → Same line. First two equal, third different → Parallel lines.

✅ 3️⃣ OPTION CHECKING METHOD (TOPPER SECRET 🤫)

MCQ gives options? DON'T solve. Put options (x,y) into BOTH equations. If both satisfy → correct answer. This is usually the fastest method in the exam!

✅ 4️⃣ ADD–SUBTRACT SPEED TRICK

If coefficients look similar (e.g., $3x+2y=10$ and $3x-2y=2$), just add them! $6x=12 \rightarrow x=2$. Instant solution 😎

✅ 5️⃣ Zero Coefficient Shortcut

If numbers look messy, try making one variable 0 mentally. It helps eliminate options fast.

🚀 EXAM SPEED STRATEGY (Golden Rule)

Question Type	Best Method
Find solution	Cross multiplication
Nature of solution	Determinant / Ratio Test
Options given	Substitute option
Similar coefficients	Add/Subtract

🧠 10-Second Memory Formula

👉 D decides destiny
👉 Options beat algebra
👉 Add when signs differ
👉 Cross multiply for direct answer

By Thunderstudy

⚡ DIFFERENTIATION: SPEED CHECKER

Class 12 Applied Maths — MCQ Shortcut Magic — By Thunderstudy

Function = Position | First Derivative = Speed | Second Derivative = Acceleration. Don't solve—spot the pattern!

🧠 SUPER FAST MCQ SHORTCUT TRICKS

✅ 1. Power Rule = Instant Answer

Shortcut:

```
d/dx(xⁿ) = n·xⁿ⁻¹
```

Hack: Reduce power by 1 and multiply by the old power. **Time saved: 20 sec**

✅ 2. Constant Killer Rule 🪄

Any constant alone disappears:

```
d/dx(7) = 0
```

MCQ Shortcut: If an option has a constant derivative ≠ 0 → ❌ **Eliminate immediately.**

✅ 3. Quotient Rule Elimination Trick

Fractions:

```
(vu' - uv') / v²
```

FAST CHECK: The denominator is **ALWAYS** squared (v^2). If an option doesn't have it → eliminate instantly.

✅ 4. Chain Rule Detector 🔗

Function inside function? Two steps: **1. Differentiate outside. 2. Multiply by derivative of inside.**

Example: $d/dx(3x+1)^5 = 5(3x+1)^4 \times 3$. The inside derivative MUST appear!

✅ 5. Maxima & Minima SUPER FAST ⭐

Step 1: Set **f'(x) = 0**. Step 2: Check f''(x):

- Positive (+) = **Minimum**
- Negative (–) = **Maximum**

💥 90% of MCQs use this rule only.

✅ 6. Tangent & Normal Instant Trick

Slope of tangent (m) = **dy/dx**. At a point? Just substitute the value. No long equation needed first.

✅ 7. MOST POWERFUL MCQ ELIMINATION TRICK 🔥

Before solving, check these four things to remove 2-3 options without calculation:

- ✔ Check Degree (Power)
- ✔ Check Sign (+/-)
- ✔ Check Denominator Square
- ✔ Check Chain Multiplication

⚡ 5-SECOND MCQ STRATEGY (Boards Topper Method)

Match pattern ≠ Solve fully.
1. Look for rule → 2. Apply pattern → 3. Match option (don't simplify!)

🧩 Example (Boards Level)

Find derivative of $\mathbf{(2x+3)^4}$.

Outside: $4(2x+3)^3$ | Inside: $\times 2$

✅ **Answer: $8(2x+3)^3$**

By Thunderstudy

🧠 INTEGRAL CALCULUS — FAST MCQ TRICKS

Class 12 Applied Maths — By Thunderstudy

Think of integration like reverse differentiation. If differentiation is breaking, integration is joining back. MCQs mostly test recognition!

⚡ 1. Recognition Trick (Most Powerful)

Instead of integrating fully, ask: **"Which function gives this when differentiated?"**

Example: $\int 2x\,dx$. Think: derivative of $x^2 = 2x$. Answer instantly → $x^2 + C$.

👉 **MCQ Secret: Options already contain the answer — just match derivative.**

⚡ 2. Formula Spotting Trick

Integral	Direct Answer
$\int x^n\,dx$	$x^{n+1}/(n+1)$
$\int 1/x\,dx$	$\ln\lvert x\rvert$
$\int e^x\,dx$	e^x
$\int \sin x\,dx$	$-\cos x$
$\int \cos x\,dx$	$\sin x$

⚡ 3. "Derivative Inside" Trick (Chain Rule Reverse)

Look for pattern:

```
∫ f'(x)·f(x) dx = (f(x))²/2
```

Example: $\int x(x^2+1)\,dx$. Derivative of (x^2+1) is $2x$. Answer: $(x^2+1)^2/2$ (approx). No expansion needed!

⚡ 4. Option Checking Method (MCQ GOLD 🥇)

Instead of solving integral: **Differentiate options.**

1. Pick an option.
2. Differentiate quickly.
3. If result matches question → DONE!

💥 **Works especially when the integral looks scary.**

⚡ 6. Definite Integral Mirror Trick

If limits are symmetric:

```
∫₋ₐᵃ f(x) dx
```

- **Odd function (x^3, sin x)** → Result = 0 ✅
- **Even function (x^2, cos x)** → Result = $2\int_0^a f(x)dx$

🔥 **Many MCQs solved in 5 seconds.**

⚡ 7. Zero Trick

```
∫ₐᵃ f(x) dx = 0
```

Same limits → Answer always 0. No thinking required!

⚡ 9. Log Substitution Pattern

If you see:

```
∫ [f'(x) / f(x)] dx
```

Answer directly: **ln |f(x)|**

🚀 5-Second MCQ Strategy

- 1️⃣ Look for formula match
- 2️⃣ Check derivative pattern
- 3️⃣ Observe limits symmetry
- 4️⃣ Differentiate options if confused

🎯 MATCH → SPOT → CHECK → WIN

By Thunderstudy

⚡ DIFFERENTIAL EQUATIONS

Shortcuts & Super Simple Tricks — By Thunderstudy

An equation that contains dy/dx (rate of change). If a normal equation tells position, a differential equation tells how it changes.

✅ TRICK 1 — Identify TYPE in 2 Seconds

Most MCQs are solved just by recognizing the type. Identification = 70% of solving.

Look at Equation	Type	Action
$dy/dx = f(x)$	Variable separable	Direct integrate
$dy/dx + Py = Q$	Linear DE	Use formula
$dy/dx = f(y/x)$	Homogeneous	Put $y=vx$
y'' present	Order = 2	Just identify

✅ TRICK 2 — Order & Degree (FREE MARKS)

- **Order:** Highest derivative.
- **Degree:** Power of the highest derivative.

Example: $(d^2y/dx^2)^3 + y = 0$. Order = 2, Degree = 3.

⚡ **Shortcut: Ignore everything except the highest derivative.**

✅ TRICK 3 — Check Option Without Solving

Instead of solving: 1️⃣ Take option 2️⃣ Differentiate it 3️⃣ Substitute into equation.

👉 **LHS = RHS? Done! Testing is faster than solving.**

✅ TRICK 4 — Linear DE Formula (MEMORISE!)

If: **dy/dx + Py = Q**

```
IF = e^∫P dx | Solution: y(IF) = ∫ Q(IF) dx
```

MCQ Hack: Just identify P and IF to match the option pattern!

✅ TRICK 5, 6 & 7 — Fast Logic

Variable Separable: If dy/dx = xy, write dy/y = x dx instantly.

General vs Particular: Contains '+ C'? General. Missing 'C'? Particular.

Constant Solution: If dy/dx = f(y), set f(y)=0 and solve for y directly.

✅ TRICK 8 — Homogeneous SPEED RULE

If every term has the same degree (e.g., dy/dx = (x+y)/x), immediately put **y = vx**. MCQs often ask for the method only!

🧠 5-SECOND MCQ STRATEGY (Exam Flow)

1. Find Type (1 sec)
2. If options given → Test solution
3. Look for formula pattern (IF)
4. Avoid full integration unless forced

🎯 **GOLDEN EXAM HACK (Topper Trick)**

"60–70% of DE questions are Recognition (Order, Degree, Type).
Try recognition first, solving later!"

By Thunderstudy

🎯 PROBABILITY DISTRIBUTION (V2)

MCQ SHORTCUT TRICKS — By Thunderstudy

Understand in 10 Seconds: A table of chances showing "What value can come?" + "How likely it is?"

⚡ TRICK 1 — The Golden Rule (Instant Check)

✅ **Sum of all probabilities = 1**

MCQ Shortcut: If missing 'k', subtract sum of others from 1.

```
k = 1 - (sum of remaining probabilities)
```

Example: 0.2 + 0.3 + k = 1 ➡️ k = 0.5. **Done in 5 seconds.**

⚡ TRICK 2 — Expected Value (Mean) SUPER FAST

Shortcut Result: Multiply row-wise → Add directly.

X	P(X)	XP(X)
1	0.2	0.2
2	0.5	1.0
3	0.3	0.9

Mean = 0.2 + 1.0 + 0.9 = 2.1

⚡ TRICK 3 — Valid Distribution Check

A table is valid ONLY IF: 1. Probabilities are 0 to 1, and 2. Total Sum = 1.

3-Second Elimination: Negative value? >1? Sum ≠ 1? ❌ **Eliminate instantly!**

⚡ TRICK 4 — Variance Without Panic 😎

Formula:

```
Var(X) = E(X²) - [E(X)]²
```

Steps: 1. Add $X^2P(X)$ column to get $E(X^2)$ | 2. Subtract Mean^2. **No long formulas.**

⚡ TRICK 5 & 6 — SD & Option Elimination Strategy

SD Hack: Find variance roughly and take the root mentally to pick the nearest option.

Topper Trick: Mean must lie between the minimum and maximum X. If X is 1-4, Mean cannot be 0 or 5!

⚡ TRICK 7 — Probability Pattern Shortcut

If probabilities increase with X (e.g., 0.1, 0.2, 0.3, 0.4), the **Mean will be towards the larger values.** Use this for estimation MCQs!

🧩 5-Step MCQ Attack Strategy

1. Check sum = 1.
2. Eliminate wrong options using Range & Sum tests.
3. Multiply X × P(X) quickly row-by-row.
4. Add column vertically for the Mean.
5. Estimate before selecting the final answer.

⏱ Average time: 20–30 seconds per MCQ

🏆 ULTRA-SHORT MEMORY FORMULA

“Multiply → Add → Square → Subtract → Root”

XP(X) → Mean → Mean^2 → Variance → SD

By Thunderstudy

🎯 BINOMIAL DISTRIBUTION

FAST MCQ TRICKS — BY THUNDERSTUDY

Think of a Coin Game: Same experiment repeated, only 2 outcomes each time (Head/Tail, Success/Failure, Pass/Fail).

✅ STEP 1: 4-Second Identification Trick

Check Questions	Condition Met?
Fixed number of trials?	✔
Only 2 outcomes?	✔
Probability same every time?	✔
Trials independent?	✔

All YES = Use Binomial Formula!

⚡ STEP 2: The Magic Formula

$$P(X=x) = {}^{n}C_{x} \cdot p^{x} \cdot q^{(n-x)}$$

Memory Trick: **"Choose × Success × Failure"**

n = total trials | x = success needed | p = success prob | q = 1 - p

🚀 STEP 3: SUPER FAST MCQ SHORTCUTS

⭐ **Trick 1 (At least one): P(at least one) = 1 - P(none)**. Always faster than summing cases!

⭐ **Trick 2 (Mean/Var):** Mean = **np** | Variance = **npq** | SD = **√(npq)**.

⭐ **Trick 3 (Symmetry):** If **p = 0.5**, then **P(X=x) = P(X=n-x)**. (Example: P(2) = P(4) if n=6).

⭐ **Trick 4 (Most Probable - Mode):** Mode = **(n+1)p**. Take the nearest integer.

🧩 STEP 4: 10-Second MCQ Strategy

1. Identify **n, p, q, x**.
2. Look for keywords: At least (Use 1-none), Average (Use np).
3. Use Symmetry (If p=0.5).
4. Apply formula only if no shortcut fits.

🏆 Example (Solved in 15 sec)

Coin tossed 3 times. Find probability of exactly 2 heads.

n=3, p=0.5, q=0.5, x=2

$$P = {}^3C_2\ (0.5)^2(0.5)^1 = 3 \times (0.5)^3 = 3/8$$

Done.

💡 ULTIMATE MEMORY LINE

"Binomial = Fixed trials + Same prob + Two outcomes"
"At least → 1 - none"

By Thunderstudy

🎯 POISSON DISTRIBUTION

MCQ MASTER SHORTCUTS — BY THUNDERSTUDY

Used for random events when the average (λ) is known.
Counting occurrences (0, 1, 2, 3...)

⭐ Main Formula & Identification

```
P(X=x) = (e⁻λ · λˣ) / x!
```

Shortcut: Find λ first. Words denoting λ: "Average", "Mean", "Rate per unit".

🔥 Trick 1: Memorize $e^{-\lambda}$ Values (80% Time Saver)

λ (Average)	$e^{-\lambda}$ (Approx)
1	0.37
2	0.14
3	0.05
4	0.018
5	0.007

🔥 Trick 2: x = 0 and x = 1 Shortcuts

No Event (x=0):

```
P(0) = e⁻λ
```

(No factorial, no power!)

Exactly One (x=1):

```
P(1) = λ · e⁻λ
```

🔥 Trick 3: Ratio Method (Biggest Time Saver ⭐)

Instead of full formula:

```
P(x+1) / P(x) = λ / (x + 1)
```

If λ=2, P(1) = 2 × P(0), P(2) = (2/2) × P(1). No formula needed, just multiply!

🔥 Trick 4: "At Least" & "At Most"

- **At least 1:** $P(X \geq 1) = 1 - P(0) = 1 - e^{-\lambda}$
- **At most 2:** $P(0) + P(1) + P(2)$
- **At least 2:** $1 - P(0) - P(1)$

🚀 10-Second MCQ Solving Strategy

1. Find **λ (Average)**.
2. Check required **x** value.
3. Start from **P(0)** using memorized values.
4. Use the **Ratio Trick** to move from P(0) to P(x).
5. Spot multiples of **$e^{-\lambda}$** in the options for an instant answer.

🧠 ULTRA MEMORY TRICK

"Zero → e power minus lambda,
Next → multiply lambda,
Then divide step by step."

By Thunderstudy

🎯 NORMAL DISTRIBUTION

FAST MCQ TRICKS — BY THUNDERSTUDY

🧠 STEP 1: The Magic Formula

No Z → No Answer! Always calculate Z-score first:

```
Z = (X - μ) / σ
```

X = value, μ = mean, σ = standard deviation

⚡ TRICK 1: Mean is the CENTER

The curve is perfectly symmetric. Area on each side of mean = **0.5**.

Direct MCQ Answer: $P(X > \mu) = 0.5$ | $P(X < \mu) = 0.5$

⚡ TRICK 2: The 68–95–99 Rule (Saves Tables!)

Range	Percentage Area
$\mu \pm 1\sigma$	68%
$\mu \pm 2\sigma$	95%
$\mu \pm 3\sigma$	99.7%

Shortcut Memory: "1–2–3 → 68–95–99"

⚡ TRICK 3: Half-Area Ladder

- Mean to 1σ = **34%**
- 1σ to 2σ = **13.5%**
- 2σ to 3σ = **2.35%**

⚡ TRICK 4 & 5: Symmetry & Complements

Negative Z: $P(Z < -a) = P(Z > a)$. Table only has positive values, use symmetry!

Greater Than: $P(X > a)$ = **$1 - P(X < a)$**. Table gives left area only.

🚀 10-Second MCQ Method

1. Identify **μ, σ, X** and find **Z**.
2. Identify type: **<** (Direct Table), **>** (1 - Table), **Between** (Big - Small).
3. Eliminate impossible options (Prob < 0 or > 1).

⭐ EXAM SECRET (Topper Trick)

Most answers lie between **0.3 and 0.9**. If you get 0.02 or 0.99, recheck your steps quickly!

Match Z-score → Spot Table Value → Pick Winner

By Thunderstudy

🎯 INFERENTIAL STATISTICS

FAST MCQ SHORTCUT TRICKS — BY THUNDERSTUDY

Population (N) = Entire Group | Sample (n) = Selected Few. Inferential statistics uses the Sample to guess about the Population.

✅ 1. Population vs Sample & Parameter vs Statistic

Memory Trick: **P** → Population → **P**arameter | **S** → Sample → **S**tatistic

Keywords: Whole/All = Population | Few/Selected = Sample

✅ 2. Hypothesis Testing - Magic Keywords

Statement Type	Identify By (Sign)
Null Hypothesis (H_0)	Equality (=, ≤, ≥)
Alternative (H_1)	Not equal (≠, <, >)

Example: Mean = 50 is H_0 | Mean ≠ 50 is H_1

✅ 3. Errors - The Courtroom Analogy ⚖️

- **Type I Error:** Rejecting H_0 when it's actually TRUE (Innocent punished).
- **Type II Error:** Accepting H_0 when it's actually FALSE (Criminal freed).

Ultra Shortcut: 1 = Reject True | 2 = Accept False

✅ 4. Level of Significance (α) & Confidence Level

Confidence Level = 1 – α

If $\alpha = 0.05$ (5%), Confidence = 95%. | If $\alpha = 0.01$ (1%), Confidence = 99%.

✅ 5. Z-test vs t-test - The Border Guard

Look only at **n** (sample size). **The border is 30!**

- **$n \geq 30$:** Large sample → **Z-test**
- **$n < 30$:** Small sample → **t-test**

✅ 6. Tailed Tests & P-Value Rule

Arrows: > or < (One direction) = **One-tailed** | ≠ (Two directions) = **Two-tailed**.

P-Value vs α: **Small p = Problem → Reject H_0** ($p \le \alpha$ means Reject H_0).

✅ 7. Estimator vs Estimate

Estimator: The Formula (e.g., $\bar{x}$) | **Estimate:** The Final Value (e.g., 52).

🚀 5-SECOND REVISION SHEET

✔ **H_0:** Equality Sign (=)
✔ **Type I:** Reject True
✔ **Type II:** Accept False
✔ **Confidence:** $1 - \alpha$
✔ **Z-test:** $n \ge 30$
✔ **Reject H_0:** p-value $\le \alpha$

THUNDERSTUDY: KEYWORDS > CALCULATIONS

⏱️ TIME-BASED DATA

Applied Maths Shortcut Guide — By Thunderstudy

When values change over time (Daily, Yearly), we use patterns to predict future values. Focus on the trend!

⚡ 2-Second Trend Direction Trick

Look only at the **First** and **Last** values:

- **Last > First:** Upward Trend 📈
- **Last < First:** Downward Trend 📉
- **Last ≈ First:** Stable / No Trend

⚡ Missing Value & Forecasting Tricks

Missing Value (Smooth Pattern): Mid-term ≈ Average of neighbors.
Example: 20, ?, 40 → ? = (20+40)/2 = 30.

Simple Forecast: **Last Value + Average Change**.

```
Average Change = (Last - First) / Number of Gaps
```

⚡ Trend Recognition: Linear vs Compound

- **Linear Trend:** If the Difference is constant (e.g., +4, +4, +4).

- **Compound/Growth:** If the Percentage (%) is constant (e.g., 100 → 110 → 121 is +10% growth).

🚀 10-Second MCQ Strategy

1. Observe pattern (Increasing? Decreasing?).
2. Check difference once.
3. If differences are equal → Linear.
4. If % is same → Compound Growth.
5. Use **Average Change** for quick predictions without a table.

Question Type	Shortcut Rule
Trend Direction	Last – First
Next Value	Last + Avg Change
Missing Value	Average of neighbors
Same Difference	Linear Trend
Same % change	Compound Growth

🎯 MATCH PATTERN → WIN MARKS

By Thunderstudy

💰 FINANCIAL MATHS

PERPETUITY | SINKING FUND | EMI — BY THUNDERSTUDY

💰 1. PERPETUITY (Money Forever)

Money paid infinitely. Keywords: "Forever", "Infinite", "Permanent".

```
Present Value = Income / Rate
```

MCQ Shortcut: Convert % to decimal (10% = 0.1) and just **DIVIDE**. Simple!

🏦 2. SINKING FUND (Future Goal)

Saving small deposits to reach a big target. Keywords: "Create Fund", "Accumulate".

$$\text{Deposit} = (\text{Future Sum} \times r) / [(1+r)^n - 1]$$

Calculator Hack: Find Numerator first. Then approximate the power (1.1^5 is approx 1.6). Pick the nearest option!

💳 3. EMI (Loan Repayment)

Monthly loan installments. **Warning:** Use <u>Monthly</u> Rate!

$$\text{EMI} = [P \cdot r \cdot (1+r)^n] / [(1+r)^n - 1]$$

Shortcut: If rate = 12% yearly, use 1% (0.01) monthly. Approximate powers to eliminate extreme options.

🚀 ULTRA-FAST IDENTIFICATION HACK

Keyword Found	Formula / Chapter
"Paid Forever"	✅ Perpetuity
"Reach Future Target"	✅ Sinking Fund
"Loan Repayment"	✅ EMI

70% of MCQs are solved just by picking the right formula!

⚡ 10-Second MCQ Strategy

1. Read **LAST line** first to find what is asked.
2. Identify keyword (Forever / Future / Loan).
3. Convert rate strictly to decimal/monthly.
4. Approximate powers for rough division.
5. Choose the nearest option.

THUNDERSTUDY: MONEY MATHS MADE EASY

📊 RETURNS, GROWTH & DEPRECIATION

MCQ FAST SOLVE GUIDE — BY THUNDERSTUDY

🔥 TRICK 1 — The Magic Multiplier

Instead of complex formulas, just use a single multiplier:

📈 **Growth (r% increase):** Multiplier = 1 + r/100 (e.g., 10% → **1.10**)

📉 **Depreciation (r% decrease):** Multiplier = 1 - r/100 (e.g., 10% → **0.90**)

```
Final Value = Initial Value × (Multiplier)ⁿ
```

⚡ TRICK 2 — Power Rule (Time Shortcut)

Don't calculate yearly. Raise the multiplier to the power of n (years).

Example: ₹100 grows at 10% for 3 years: $100 \times (1.1)^3 = 133.1$. Done!

⚡ TRICK 3 & 4 — The Rule of 72 (MCQ Weapon)

To find when an amount DOUBLES (Growth) or HALVES (Depreciation):

```
Years ≈ 72 / Rate
```

Example: Rate = 12%. Years = 72 / 12 = 6 years. Instant Answer.

⚡ TRICK 5 — Successive Changes Cheat

If a value increases 10% then decreases 10%, it's NOT the same!

(1.10) × (0.90) = 0.99 (Final is 99% of original — always a loss).

⚡ Mental Multiplier Table

Change	Multiplier	Change	Multiplier
5% Increase	× 1.05	10% Decrease	× 0.90
10% Increase	× 1.10	20% Decrease	× 0.80
25% Increase	× 1.25	5% Decrease	× 0.95

🚀 10-Second MCQ Flow

1. Identify if it is **Growth** or **Depreciation**.
2. Convert % into the **Multiplier**.
3. Raise to the **Power** (years).
4. Multiply once and match the closest option.
5. **Eliminate:** If Growth, remove smaller options; if Depreciation, remove larger ones.

⭐ MEMORY LINE (Revise Before Exam)

"Plus means grow, minus means slow —
Make multiplier, power it, answer will show."

THUNDERSTUDY: CALCULATE LESS, THINK MORE

⚡ LINEAR PROGRAMMING (LPP)

MCQ SHORTCUT TRICKS — BY THUNDERSTUDY

🚀 TRICK 1 — Corner Point Rule (The Golden Law)

The optimal solution (Max or Min) **ALWAYS** lives at a corner point (vertex).

MCQ Hack: Don't draw the graph. Substitute the corner points from options directly into Z!

🚀 TRICK 2 — Inequality & Shading (2-Sec Rule)

- **≤ (Less than or equal):** Shade Towards origin (0,0).
- **≥ (Greater than or equal):** Shade Away from origin.

Quick Check: Put (0,0) into the equation. If it works, origin is included!

🚀 TRICK 3 — Infinite vs No Solution

Infinite Solutions: If the Objective Function (Z) has the same coefficients as a constraint (parallel lines).

No Solution: If constraints are opposite (e.g., $x+y \geq 10$ and $x+y \leq 2$). Impossibility = No region.

🚀 TRICK 4 — Fast Option Elimination

1. If **$x, y \geq 0$**, immediately reject any option with negative numbers ❌
2. Test if the point satisfies all constraints. If it fails even one, reject it!
3. Substitute remaining valid points into **Z**. Choose the winner.

🚀 TRICK 5 — Quick Substitution Example

Maximize: $Z = 3x + 2y$ | Points: (0,0), (2,0), (1,3), (3,1)

Point	Calculation	Result
(0,0)	3(0) + 2(0)	0
(2,0)	3(2) + 2(0)	6
(1,3)	3(1) + 2(3)	9
(3,1)	3(3) + 2(1)	**11 (Winner)**

🏆 BOARD EXAM SPEED STRATEGY

"Skip the graph, test the points, find the corners."

- 1. Read Z first.
- 2. Check provided vertices in options.
- 3. Validate against constraints.
- 4. Pick Max/Min.

THUNDERSTUDY: NO GRAPH, NO STRESS